



Solar skylights for your roof

Researchers at the University of Exeter created a thick block of glass with optical features that works with solar power. <http://digit.in/Lightsolar>



Aston Martin luxury boat

Aston Martin launches a \$1.7 million boat equipped with a champagne cooler, convertible sofa and many other fancy features. <http://digit.in/MartinBoat>

regions, enabling them to build resistance to the new cold climate. According to them, this was confirmed by DNA markers, gene expression levels, and physiology measures.

Most of the reported instances of rapid evolution have been linked to human intervention either through natural habitat encroachment or climate change (pollution to be precise). Pollution is already considered a big factor forcing rapid evolution and another instance was observed in the State of New Jersey, US. The Atlantic killifish thriving in the Lower Passaic River were burdened with toxic leftovers including DDT. Surprisingly, few pockets of the killifish population were able to adapt to the toxic waters and managed to survive. A specific water flea called *Daphnia* was found to have an increased thermal tolerance in just a duration of two years, owing to increasing temperatures thanks to global warming. To understand rapid evolution better, biologists wanted to study their causes better and try to repeat them.

STUDYING RAPID EVOLUTION

The best way to study rapid evolution is to observe it in a natural habitat rather than depend only on controlled lab experiments. We already talked about how microorganisms or smaller organisms are favourable to study because of their shorter lifespans and rapid reproductive cycle. Hence, it is much easier to study or observe mutations or any evolutionary changes in them. Biologists have also ventured into isolated regions or islands to introduce an external factor or species to influence evolution in that area. They weren't certain whether they would observe dramatic evolutionary changes or even finish the study in their lifetime. But they were in for a surprise.

A lab study by the University of Leeds found that the age of adulthood of soil mites doubled in just 15 generations and there was also a significant change in population sizes. Soil mites were taken from the wild and transferred into 18 glass tubes. While they

reproduced in the controlled environment, forty percent of adult mites were removed every week from six of the tubes. Similarly, juveniles mites were removed from six other tubes. The remaining six tubes were left untouched. The consequence of these conditions led to a slow growth in juveniles in all but the six untouched tubes. And when they finally matured to reproduce, they were larger compared to previous generations. This further made it easy for them to lay eggs, highly influencing population sizes.

Forcing species into new habitats can either completely shuffle the ecosystem or they might simply die out. An island experiment with Italian wall lizards was conducted in 1971, where they were moved to a neighbouring island in the south Adriatic Sea. Wall lizards from Pod Kopiste were introduced to a neighbouring tiny island called Pod Mrcaru, and after a few decades, scientists witnessed shocking results. It wasn't until 2004 that the team of researchers went back to the island and found that the island was swarming with a new species of lizards. However, these lizards had evolved drastically, something which the researchers say would have taken millions of years. The lizards developed a completely new gut structure to accommodate a vegetarian diet which was originally built for an insect-based diet. Their expanded gut was able to digest the vegetation's cellulose turning them into fatty acids. They also developed a larger head to support stronger bites to munch on the plants. Equipped with a plant-based diet, the lizards didn't have to depend on insects for food and could feed on plants abundantly present around them. Because of this, survival wasn't much of a problem and the species flourished to grow in population.

Back in 1978, biologist David Reznick (then a graduate student) had set out on a journey to Trinidad to "watch evolution happen". The test subjects were guppies, a species of tropical fish in the Caribbean islands with a high reproductive drive. Reznick wanted to experiment whether preda-

tors could affect the genetic adaptation of guppies in a controlled environment. He moved a batch of guppies from a location rich with cichlids, a species of fish who feed on them, to a stream without any predators. Alternately, predator-free locations with guppies were populated with cichlids. After going back to these sites, Reznick observed remarkable changes where the predator-rich areas had smaller-sized guppies who matured at younger ages. This was driven by the aggressive predators that made more production of babies a priority. Whereas, the predator-free areas saw larger guppies maturing at a later age and reproducing slowly. Their survival wasn't at stake anymore, hence they evolved to take things slowly. And the most shocking observation was that these changes occurred in just a span of four years.

CONSEQUENCES

It's clear that humans had a significant role to play in the recently observed instances of rapid evolution in nature. Of course, we can't say for sure whether these changes are going to last in those species over the long run. Not just global warming or climate change, we have other activities such as hunting, harvesting, fishing, agriculture, urbanisation, and a number of other causes responsible of these changes. We have not only affected the global ecology of the planet, but as mentioned above, we are interfering with the evolutionary trajectories of several species. In a human-dominated environment, this is bound to happen and it's unlikely that our interference with the natural order of things are going to reduce. The next best step forward is to understand the trajectory that we have induced on these species to predict how things are going to turn out in the future. We also need to have enough data to be able to predict the economic and social costs of these changes, in order to prepare ourselves for the implied consequences. It would be an ironic example of karmic justice if our influence on the global ecology comes back to threaten our very existence on this planet. 